CSA04-Operating Systems

TASK 3 – Evaluate Operating-System Structures

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1. Aim

To evaluate different operating-system structures based on reliability, performance and ease of updating, and select suitable structures for patient monitors and the hospital database server.

2. Comparison of Operating-System Structures

OS structure	Explanation	Reliability	Performance	Ease of updating
Monolithic	All major OS services run inside one kernel.	Lower fault isolation	High	Difficult
Layered	OS is divided into different layers.	Good	Moderate	Easy
Microkernel	Only essential services run in the kernel; other services run separately.	High fault isolation	May have communication overhead	Easy
Modular	Kernel supports loadable modules.	Good, but faulty modules can affect the kernel	High	Easy
Hybrid	Combines features of monolithic and microkernel structures.	Good	High	Moderate

3. Suitable Structures for Hospital Environments

A. Patient Monitoring System – Microkernel

- Provides better isolation between OS services.
- Reduces the amount of code running in kernel mode.
- Improves reliability if a service fails.
- Supports real-time scheduling in suitable RTOS implementations.
- Helps protect critical monitoring operations.

Justification: A microkernel-based RTOS such as QNX is suitable because patient monitors require reliable and predictable alarm responses within 10 milliseconds.

B. Database Server – Modular Structure

- Supports loadable kernel modules.
- Provides high performance.
- Allows device drivers and features to be updated separately.
- Supports efficient resource management.
- Works well with Linux-based clustered servers.

Justification: A modular Linux kernel is suitable for the database server because it provides performance, flexibility and easier maintenance.

4. Structures Used in Popular Operating Systems

Operating system	Structure	Explanation
Linux	Monolithic with modular support	Core services run in the kernel, with loadable modules.
Windows	Hybrid	Combines microkernel and monolithic design features.
Android	Layered architecture with Linux kernel	Uses the Linux kernel, hardware abstraction and application framework.
FreeRTOS	Small real-time kernel	Provides predictable task scheduling with a small memory footprint.
QNX	Microkernel	Runs many services and drivers outside the kernel for better isolation.

5. Alternative Concept and Complexity Analysis

Alternative: Use a monolithic kernel instead of a microkernel for bedside patient monitors.

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Time Complexity

- Monolithic: A system call involves switching to kernel mode and executing the service. Its overhead is generally lower.
- Microkernel: A service request may involve message passing, context switches and communication between servers, increasing overhead.

Space Complexity

- Microkernel: Uses less kernel memory because many services run outside the kernel.
- Monolithic: Requires more kernel memory because most OS services and drivers run inside the kernel.

Effect of a Faulty Driver

- In a monolithic kernel, a faulty driver can crash the entire operating system.
- In a microkernel, a separate driver service can fail without necessarily affecting the entire system.

Evaluation: Although a monolithic kernel may provide faster service requests, a microkernel is preferable for patient monitors because reliability and fault isolation are more important than small communication overheads. The complete system must still be tested to ensure the 10 ms deadline is met.

6. Final Evaluation

Patient monitors	Microkernel	Reliability and fault isolation
Database server	Modular	Performance and easy maintenance
Doctors' desktops	Hybrid	Multitasking and flexibility
Nurses' tablets	Layered	Organized and manageable design
Imaging server	Modular	Efficient processor and hardware support

7. Result

Thus, different operating-system structures were evaluated based on reliability, performance and maintainability. A microkernel-based RTOS is recommended for bedside patient monitors, while a modular Linux structure is recommended for the database server to ensure efficient and reliable hospital operations.